Multiple Regression Analysis: Data Scaling, Functional Forms, Selection of Regressors, Prediction

These notes accompany Lecture 4. They combine the slide material with the explanations given in class. Terms printed in bold are defined in the glossary at the end.

These topics are not fundamental in the way that estimation and inference are – but they are what makes multiple regression usable across a broad range of empirical problems.

Learning outcomes – after this lecture you should be able to:

1. Produce a point prediction from an estimated equation, and explain the difference between a confidence interval for the mean and a prediction interval for an individual outcome.
2. Explain why exponentiating a predicted log value underpredicts y, and correct for it using either the normality-based adjustment or the smearing estimate.
3. Convert between approximate and exact percentage-change interpretations of a log-level coefficient.
4. Interpret a quadratic specification, compute the marginal effect at a given point, and locate the turning point.
5. Interpret an interaction term, and explain why centring the variables makes the other coefficients meaningful.
6. Compare competing models using adjusted R-squared, AIC and SBC, and explain what each penalises.

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1. Prediction

1.1 Point prediction

Once a model is estimated, plugging particular values of the regressors into the equation gives a prediction for y – an estimate of the expected value of y given those values of the explanatory variables.

Take the CEO salary equation:

$$\textbf{salary} = 613.44 + 0.019\,\textbf{sales} + 0.023\,\textbf{mktval} + 12.703\,\textbf{ceoten}$$

where salary is CEO salary (in $ thousands), sales is firm sales (in $ millions), mktval is market value of the firm (in $ millions), and ceoten is years as CEO with the current company.

To predict salary for sales = 5 000 ($5 billion), mktval = 10 000 ($10 billion) and ceoten = 10:

$$E(\text{salary} \mid \text{sales} = 5000, \text{mktval} = 10000, \text{ceoten} = 10) \approx 1\,069.6$$

that is, about $1.07 million. If you compute this by hand from the rounded coefficients printed above you will get roughly 1 065 – the small discrepancy is entirely due to rounding in the displayed coefficients, and EViews uses the unrounded values.

Doing it in EViews

1. Extend the sample range and add a new observation (here observation 178).
2. Enter the chosen values of sales, mktval and ceoten for that observation.
3. From the estimated equation window choose Proc → Forecast.
4. Specify a name for the forecast series, a name for the standard error series (needed for the interval), and the forecast sample.

1.2 Two kinds of interval – keep them apart

EViews will draw a band around the forecast. It is worth being precise about what that band represents, because two different questions are easily confused:

Question	Interval	What varies
Where is the average salary of all CEOs with these characteristics?	Confidence interval for E(y | x)	Only sampling error in the estimated coefficients
What salary will this particular CEO earn?	Prediction interval for an individual y	Coefficient sampling error plus the error term u

The prediction interval is always the wider of the two, because an individual outcome carries its own unobserved u on top of the uncertainty about where the regression line sits. The band EViews draws from the forecast standard error is a prediction interval for an individual outcome – so if what you want is an interval for the conditional mean, the EViews band is too wide.

The ±2 standard error band is an approximate 95% interval; the exact multiplier is the relevant t critical value, which is close to 2 in any reasonably large sample.

1.3 Predicting y when the dependent variable is in logs

Because the log transformation is so common for the dependent variable, this case deserves separate treatment. Given the OLS estimates we can predict log(y) for any values of the regressors:

$$\widehat{logy} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \cdots + \hat{\beta}_k x_k$$

Since the exponential undoes the log, the obvious guess is simply to exponentiate:

$$\hat{y} = exp(\widehat{logy}) \quad \leftarrow \textit{naïve prediction}$$

This does not work. It systematically underestimates the expected value of y – and EViews does not correct for it automatically.

> **Why the naïve prediction is biased downward**
>
> Positive and negative log-errors are not symmetric once you exponentiate them.
>
> A large positive error raises y by more than an equally large negative error lowers it – upside effects exceed downside effects.
>
> So the variability in the log-errors does not cancel out; it creates a positive contribution to E(y) that exponentiating a point prediction ignores.
>
> Ignoring this asymmetry therefore underpredicts y systematically – not occasionally, and not in a way that averages out over observations.

Two corrections

Under the full CLM assumptions (MLR.1–MLR.6, so with normal errors), the consistent prediction multiplies the naïve one by a constant determined by the standard error of the regression:

$$\hat{y} = \exp(\hat{\sigma}^2/2)\exp(\widehat{logy})$$

This relies on normality of u, which we would rather not require. Without it, replace the theoretical constant by its sample analogue – the average of the exponentiated residuals:

$$\hat{y} = exp(\widehat{logy}) * \frac{1}{n}\sum_{i=1}^{n} exp(\hat{u}_i) \quad \leftarrow \textit{smearing estimate}$$

Method	Multiplier applied to exp(predicted log y)	Requires normality?
Naïve	1 (no correction) – biased downward	–
Normality-based	$\exp(\hat{\sigma}^2/2)$	Yes
Smearing estimate	$(1/n)\ \Sigma \exp(\hat{u}_i)$	No

Worked example in EViews

Estimating the salary model in log form, which is the more adequate specification:

log(salary) = 4.50 + 0.163 log(sales) + 0.109 log(mktval) + 0.012 ceoten

EViews produces the naïve forecast. To correct it, first save the residuals as a series, then in the command window:

```
scalar smear = @mean(exp(resid01))
series yhat = exp(salaryflog) * smear
```

The resulting predicted salary is about $1 263 059.

> **One practical point**
>
> R-squared: in a model with log(y) as dependent variable, R^2 is the fraction of the variation in log(y) explained – not the variation in y itself. To get the latter, compute the squared correlation between the actual y and the exponentiated fitted values.

2. Transformations of variables

Logarithms are the most popular device in econometrics for allowing non-linear relationships between the explained and explanatory variables. This section covers the main variations used in applied work: logs, quadratics and interactions.

2.1 Interpreting parameters in a log model

For a sample of 506 communities in the Boston area, relate median housing price to nitrogen oxide in the air (a pollution measure) and the average number of rooms per house:

$$log(price) = \beta_0 + \beta_1 log(nox) + \beta_2 rooms + u$$

- β_1 is the **elasticity** of price with respect to nox – both variables are in logs.
- β_2 is the change in log(price) when rooms rises by one. Multiplied by 100 it is the approximate percentage change in price; $100 \cdot \beta_2$ is the **semi-elasticity** of price with respect to rooms.

Estimated on the data:

$$\mathbf{log(price) = 9.234 - 0.718\ log(nox) + 0.306\ rooms}$$

- When nox increases by 1%, price falls by 0.718%, holding only rooms fixed.
- When rooms increases by one, price increases by approximately 100 × 0.306 = 30.6%.

That second figure is somewhat inaccurate, and it is worth understanding why.

2.2 Exact versus approximate percentage change

The interpretation "multiply by 100" rests on the approximation $\%\Delta y \approx 100 \cdot \Delta log(y)$, which is good for small changes and deteriorates as $\Delta log(y)$ grows. A change of 0.306 in a log is not small.

Simple properties of exponentials and logs give the exact percentage change in the predicted y:

$$\%\Delta\hat{y} = 100 * \left[exp\left(\hat{\beta}_2 \Delta x_2\right) - 1\right]$$

Applied to the housing example in both directions:

Change in rooms	Exact calculation	Result
$\Delta x_2 = +1$	$100 \cdot [\exp(0.306) - 1]$	+35.8%
$\Delta x_2 = -1$	$100 \cdot [\exp(-0.306) - 1]$	−26.4%
Approximation (either direction)	$100 \cdot 0.306$	±30.6%

Change in rooms	Exact calculation	Result

The pattern worth remembering

Adding a room raises the predicted price by 35.8%; removing one lowers it by 26.4%. The two are not mirror images – the exact effect is asymmetric.

The simple approximation, 30.6%, always falls between the absolute values of the two exact figures. So it is never badly wrong, but it is never exactly right either, and it hides the asymmetry entirely.

2.3 Considerations when using logarithms

- Coefficients have percentage-change interpretations, and changing the units of a variable has no effect on its coefficient when that variable appears in logs.
- Logs are commonly used for amounts that are always positive (GDP, wage, population, sales). They are used less often for variables measured in years (education, age, experience), and infrequently for variables that are already percentages or proportions (unemployment or interest rates).
- Models with log(y) as the dependent variable often satisfy the CLM assumptions more closely: linearity has a better chance of holding, homoskedasticity is more likely, and normality is often more plausible.
- For large changes in an explanatory variable, use the exact formula of Section 2.2 rather than the approximation.

Two limitations of logs

The log cannot be used at all if the variable takes zero or negative values.

Having the dependent variable in logs makes it harder to predict the original variable – which is precisely the problem Section 1.3 was solving.

2.4 Quadratics: decreasing or increasing effects

Quadratic functions are used frequently in applied economics to capture marginal effects that change with the level of the variable:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + u$$

The essential point: β_1 no longer measures the change in y with respect to x. Both terms are involved:

$$\Delta\hat{y}/\Delta x \approx \hat{\beta}_1 + 2\hat{\beta}_2 x$$

The marginal effect now depends on where you evaluate it. Setting x = 0 shows that $\hat{\beta}_1$ can be read as the approximate slope in going from x = 0 to x = 1 – which is why the coefficient on x in a quadratic model measures the partial effect starting from zero.

Returns to experience

With y = wage and x = exper, the estimated relationship has $\hat{\beta}_1 > 0$ and $\hat{\beta}_2 < 0$, implying that experience has a diminishing effect on wage:

$$\widehat{wage} = 3.73 + .298\, exper - .0061\, exper^2$$
$$(.35)\ \ (.041)\qquad\quad (.0009)$$
$$n = 526,\ R^2 = .093.$$

	Calculation	Value
First year of experience	$\hat{\beta}_1 = 0.298$	≈ 30¢ per hour
Second year of experience	0.298 − 2(0.0061)(1) = 0.286	≈ 28.6¢ per hour

Each additional year is worth slightly less than the one before. Eventually the marginal effect turns negative, at the turning point:

$$x^* \approx -\hat{\beta}_1/(2\hat{\beta}_2)$$

Before x*, x has a positive effect on y; after it, the effect is negative. Familiar examples of this shape include the Laffer curve relating tax rates to government revenue.

2.5 Considerations when using quadratics

- A quadratic allows the effect of x to increase or decrease with the level of x.
- The turning point is easy to compute, and you should always compute it – to check whether it falls inside the range of the data and whether it makes economic sense. A turning point at 60 years of experience, in a sample topping out at 40, is not a finding about the world.
- If the two coefficients have opposite signs the turning point is strictly positive; if they share a sign, the turning point sits at a negative value of x and so is usually irrelevant.
- A seemingly tiny coefficient on the squared term can be practically important, because it accumulates: what it implies is a changing slope, not a small effect.
- The coefficient on x measures the partial effect starting from x = 0.

> **The other shape**
>
> When $\beta_1 < 0$ and $\beta_2 > 0$ the quadratic is U-shaped rather than parabolic, capturing an effect of x on y that increases with x.
>
> Average cost curves are the standard example: increasing returns to scale at first, as fixed costs are spread over a larger output, then a point of optimal efficiency, then decreasing returns.

2.6 Interaction terms

Sometimes the partial effect of one explanatory variable on y depends on the magnitude of another explanatory variable. Consider a house price model:

$$price = \beta_0 + \beta_1 sqrft + \beta_2 bdrms + \beta_3 sqrft \cdot bdrms + \beta_4 bthrms + u$$

The partial effect of bedrooms on price, holding everything else fixed, is now:

$$\Delta price/\Delta bdrms = \beta_2 + \beta_3\, sqrft$$

If $\beta_3 > 0$, an additional bedroom adds more to the price of a larger house – there is an interaction effect between square footage and number of bedrooms.

The interpretation trap

In this equation β_2 is the effect of bdrms on price for a home with zero square feet.

That is not a quantity anyone cares about, and no such house exists in the data. Its estimate will typically be insignificant – and if you do not notice why, you may wrongly conclude that bedrooms do not affect price.

2.7 Centring the variables

The fix is to reparameterise so the original coefficients mean something. For a model with two regressors and their interaction, subtract the means before forming the interaction term:

$$y = \delta_0 + \delta_1 x_1 + \delta_2 x_2 + \beta_3 (x_1 - \mu_1)(x_2 - \mu_2) + u$$

where μ_1 and μ_2 are the population means. Now δ_2 is the partial effect of x_2 on y at the mean value of x_1, and δ_1 has the corresponding interpretation. Subtracting the means – or any other interesting values, such as the median or the quartiles – gives the coefficients on the original variables a useful reading.

EViews example

Regressing house prices on a location indicator (nearinc), area, and number of bathrooms, with a centred interaction between area and baths:

$$log(price) = \beta_0 + \beta_1 nearinc + \beta_2 area + \beta_3 baths + \beta_4 (baths - \bar{b}aths)(area - \bar{a}area) + u$$

The coefficient on the interaction is statistically significant, supporting the hypothesis that an additional bathroom adds more to the price of a larger house.

Centred model

Dependent Variable: LOG(PRICE)
Method: Least Squares
Date: 01/03/26 Time: 17:08
Sample: 1 321 IF YEAR=1978
Included observations: 179

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	10.37039	0.072191	143.6517	0.0000
NEARINC	-0.178264	0.042320	-4.212294	0.0000
AREA/1000	0.177305	0.037380	4.743333	0.0000
BATHS	0.210765	0.034279	6.148487	0.0000
(AREA/1000-@MEAN(AREA/1000,"1 321 if ...	0.077750	0.031187	2.493042	0.0136

R-squared	0.599203	Mean dependent var	11.17908
Adjusted R-squared	0.589989	S.D. dependent var	0.366974
S.E. of regression	0.234981	Akaike info criterion	-0.031089
Sum squared resid	9.607597	Schwarz criterion	0.057944
Log likelihood	7.782439	Hannan-Quinn criter.	0.005014
F-statistic	65.03379	Durbin-Watson stat	1.917542
Prob(F-statistic)	0.000000		

Uncentred model

Dependent Variable: LOG(PRICE)
Method: Least Squares
Date: 03/20/26 Time: 12:16
Sample: 1 321 IF YEAR=1978
Included observations: 179

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	10.72910	0.156235	68.67267	0.0000
NEARINC	-0.178264	0.042320	-4.212294	0.0000
AREA/1000	-0.002085	0.087021	-0.023956	0.9809
BATHS	0.055294	0.066652	0.829585	0.4079
(AREA/1000)*(BATHS)	0.077750	0.031187	2.493042	0.0136

R-squared	0.599203	Mean dependent var	11.17908
Adjusted R-squared	0.589989	S.D. dependent var	0.366974
S.E. of regression	0.234981	Akaike info criterion	-0.031089
Sum squared resid	9.607597	Schwarz criterion	0.057944
Log likelihood	7.782439	Hannan-Quinn criter.	0.005014
F-statistic	65.03379	Durbin-Watson stat	1.917542
Prob(F-statistic)	0.000000		

The key comparison – and the reason centring matters

The centred and uncentred models are statistically identical: same R-squared, same fit, same summary statistics. Centring changes no substantive result.

What changes is the interpretation. In the uncentred model, the coefficients on area and baths give partial effects at zero baths and zero area respectively – and since no such houses exist in the sample, both come out insignificant.

In the centred model the same coefficients give partial effects at mean values, and are informative.

The lesson: an insignificant coefficient in a model with interactions may simply mean you have evaluated the effect somewhere meaningless.

2.8 Considerations when using interactions

- Interaction terms let the partial effect of x_1 depend on the level of x_2, and vice versa – the relationship is symmetric.
- Interpretation is tricky: β_1 measures the partial effect of x_1 when $x_2 = 0$, which may be impossible or uninteresting. Centring both variables around interesting values before constructing the interaction gives a more readable equation.
- A standard t-test determines whether an interaction term is statistically significant. To judge its practical importance, compute the partial effects at several values of the other variable.

3. Selection of regressors

3.1 Adjusted R-squared

R^2 can never fall when a new independent variable is added, which makes it useless for deciding whether a variable belongs in the model. The adjusted R-squared imposes a penalty for additional regressors:

$$\bar{R}^2 = 1 - (1 - R^2) \cdot (n - 1)/(n - k - 1)$$

The penalty has a precise consequence worth memorising:

> **When does adjusted R^2 rise?**
>
> Adding a variable raises adjusted R^2 if and only if the t-statistic on the new variable exceeds one in absolute value.
>
> So adjusted R^2 can fall when a variable is added – which is exactly what makes it usable for comparing models with different numbers of regressors.
>
> Note the threshold, though: $|t| > 1$ is a far weaker standard than $|t| > 2$. A variable can raise adjusted R^2 while remaining statistically insignificant at conventional levels.

Adjusted R^2 can also be negative, which happens when the model fits worse than simply using the sample mean of y. That is a signal to reconsider the specification.

One further reminder from Lecture 2: a low R^2 does not make a regression useless. Even when the regressors collectively explain little of the variation in y, the OLS estimates may still be reliable estimates of the partial effects.

3.2 Information criteria

An alternative approach uses explicit model selection criteria. The two most common:

$$AIC = n \cdot \ln(SSR) + 2k \qquad \text{(Akaike Information Criterion)}$$

$$SBC = n \cdot \ln(SSR) + k \cdot \ln(n) \qquad \text{(Schwarz Bayesian Criterion)}$$

Both trade fit against parsimony: the first term falls as the fit improves, the second rises with the number of parameters. We prefer the model that minimises them.

Note that SBC's penalty, $k \cdot \ln(n)$, exceeds AIC's $2k$ whenever $n > 7$ – so SBC is the more conservative criterion and tends to select smaller models. It is common for the two to disagree; a model that outperforms on several criteria is generally preferred, and in practice AIC is the most widely used.

> **A practical warning about the numbers**
>
> The formulas above and the AIC/SBC values EViews reports are on different scales – EViews reports a log-likelihood-based version divided by n, so its figures are usually small numbers, while the formulas above give bigger values.
>
> The two orderings of models coincide, so your conclusions will be the same. But do not expect the number you compute by hand to match the number on the screen.

3.3 A worked comparison

Three competing specifications estimated on the same 353 observations:

	SSR	R^2	Adjusted R^2	AIC	SBC
Model 1 (3 regressors)	185.31	0.6235	0.6202	2.2161	2.2599
Model 2 (3 regressors)	183.86	0.6264	0.6232	2.2083	2.2521
Model 3 (4 regressors)	183.60	0.6270	0.6227	2.2125	2.2673

Read the table across, and the logic of model selection becomes visible:

- **Model 3 has the highest R^2** and the lowest SSR – it fits the sample best, as it must, since it has the most regressors.
- **But Model 3 has a lower adjusted R^2 than Model 2,** and worse AIC and SBC. The extra variable does not earn its keep: it improves the fit by less than the penalty for adding it.
- **All three criteria agree on Model 2.** When adjusted R^2, AIC and SBC point the same way, the choice is easy – and this is the ordinary case, not a lucky one.

This is precisely why R^2 alone cannot be used to choose a model, and why the penalised criteria exist.

4. Summary

1. A point prediction is obtained by plugging chosen values of the regressors into the estimated equation.
2. Exponentiating a predicted log value underpredicts y systematically, because log-errors are asymmetric after exponentiation. Correct with $\exp(\hat{\sigma}^2/2)$ under normality, or with the smearing estimate – the average of the exponentiated residuals – without it.
3. In a log-level model, $100 \cdot \hat{\beta}$ is an approximate percentage change. The exact figure is $100 \cdot [\exp(\hat{\beta}\Delta x) - 1]$, and it differs for increases and decreases; the approximation always lies between the two exact magnitudes.
4. A quadratic makes the marginal effect depend on the level: $\Delta\hat{y}/\Delta x \approx \hat{\beta}_1 + 2\hat{\beta}_2 x$. Always compute the turning point $-\hat{\beta}_1/(2\hat{\beta}_2)$ and check whether it lies inside the data.

5. An interaction term makes the partial effect of one variable depend on another. Coefficients on the uncentred variables give effects at zero, which is often meaningless; centring makes them effects at the mean without changing the fit.
6. R^2 cannot fall when a regressor is added, so use adjusted R^2, AIC or SBC to compare models of different sizes. Adjusted R^2 rises whenever $|t| > 1$ – a weaker standard than statistical significance.

> **The theme running through this lecture**
>
> Every technique here is about making a coefficient answer the question you actually asked.
>
> Exponentiate carelessly and you predict the wrong thing. Read a log coefficient as exact and you overstate a decrease. Leave an interaction uncentred and you evaluate an effect at a house with zero square feet. Choose on R^2 alone and you always pick the biggest model.
>
> None of these are estimation failures. The estimates are fine; the interpretation is where the work is.

Key terms

Term	Meaning
Point prediction	The fitted value obtained by plugging chosen regressor values into the estimated equation; an estimate of $E(y \mid x)$.
Confidence interval for $E(y\|x)$	Interval for the conditional mean; reflects sampling error in the coefficients only.
Prediction interval	Interval for an individual outcome; reflects coefficient uncertainty plus the error term. Always wider.
Naïve prediction	exp of the predicted log(y). Systematically underestimates E(y).
Smearing estimate	$(1/n)\,\Sigma \exp(\hat{u}_i)$. Multiplicative correction to the naïve prediction that does not require normality.
Elasticity	Percentage change in y from a 1% change in x; the coefficient in a log-log specification.
Semi-elasticity	Percentage change in y from a one-unit change in x; $100\cdot\beta$ in a log-level specification.
Exact percentage change	$100\cdot[\exp(\hat{\beta}\Delta x) - 1]$. Differs for increases and decreases, unlike the approximation.
Quadratic specification	Inclusion of x and x^2, making the marginal effect $\hat{\beta}_1 + 2\hat{\beta}_2 x$ depend on the level of x.
Turning point	$x^* \approx -\hat{\beta}_1/(2\hat{\beta}_2)$. Where the marginal effect changes sign.
Interaction term	A product of two regressors, letting the partial effect of each depend on the level of the other.
Centring	Subtracting means (or other interesting values) before forming an interaction, so the other coefficients give effects at those values.
Adjusted R-squared	$1 - (1-R^2)(n-1)/(n-k-1)$. Rises when a variable is added if and only if $\|t\| > 1$.
AIC	$n\cdot\ln(SSR) + 2k$. Penalises additional parameters; minimise.
SBC (Schwarz)	$n\cdot\ln(SSR) + k\cdot\ln(n)$. A heavier penalty than AIC for $n > 7$; favours smaller models.

Review questions

Use these to check your understanding before the seminar.

1. Using salary = 613.44 + 0.019 sales + 0.023 mktval + 12.703 ceoten, predict salary for sales = 8 000, mktval = 4 000, ceoten = 5. State the units of your answer.
2. Explain in your own words why exp(predicted log y) underestimates E(y). Why does the bias not average out across observations?
3. Contrast the two corrections for log-dependent-variable prediction. Under what circumstance would you prefer the smearing estimate, and what do you give up by using $\exp(\hat{\sigma}^2/2)$ instead?
4. A colleague reports $R^2 = 0.72$ for a model with log(wage) as the dependent variable and says "the model explains 72% of the variation in wages". What is wrong with this, and how would you compute the figure they intend?
5. In log(price) = 9.234 − 0.718 log(nox) + 0.306 rooms, compute the exact percentage change in price from (a) adding one room, (b) removing one room. Explain why they are not equal in magnitude.
6. Would you expect the approximation to be better or worse if $\hat{\beta}$ were 0.05 instead of 0.306?
7. For which of these variables would you normally take logs, and why: GDP; the unemployment rate; years of schooling; firm sales; a firm's net profit?
8. In a wage-experience quadratic with $\hat{\beta}_1 = 0.298$ and $\hat{\beta}_2 = -0.0061$, compute the marginal effect of experience at 5 years and at 30 years. Then find the turning point and comment on whether it is economically sensible.
9. A researcher reports a coefficient of −0.0002 on x^2 and dismisses it as "too small to matter". Explain why this reasoning is unsound.
10. In price = $\beta_0 + \beta_1$sqrft + $\beta_2$bdrms + $\beta_3$sqrft·bdrms + u, write the partial effect of bdrms. What does β_2 measure, and why is it usually uninteresting?
11. The centred and uncentred interaction models have identical R^2. Explain what centring does and does not change, and why an insignificant coefficient in the uncentred version is not evidence that the variable has no effect.
12. A variable with t = 1.4 is added to a model. What happens to R^2, to adjusted R^2, and would you keep the variable? Justify your answer.
13. Using the three-model table in Section 3.3, explain why Model 3 has the best R^2 but is not the preferred model. Which criterion would you report, and why?
14. AIC and SBC disagree about two models. Explain the source of the disagreement in terms of their penalty terms, and state which is more likely to favour the larger model.

References and further reading

Wooldridge, J. M. Introductory Econometrics: A Modern Approach – Chapter 6 covers the material of this lecture.